Employee Learning & Skill Certification Tracker (LSCT)

Load Test Report

1. Introduction

The purpose of this load test was to evaluate the performance, stability, and scalability of the Employee Learning & Skill Certification Tracker (LSCT) application under simulated concurrent user activity. The test focused on the Employee Courses API to verify that the system can handle expected traffic while maintaining acceptable response times and reliability.

2. Test Configuration

The load test was performed using Artillery, an open-source load testing tool. The test simulated 50 concurrent virtual users over a duration of 60 seconds, targeting the Employee Courses API. A total of 3000 requests were generated during the test to assess how the API performs under sustained load.

3. Performance Results

The table below summarizes the key performance metrics recorded during the load test.

Metric	Value
Requests per Second	50 req/sec
Total Requests	3000
Successful Requests	3000
Failed Requests	0
Error Rate	0%
Mean Response Time	89 ms
Median (P50)	85.6 ms
P95 Latency	106.7 ms
P99 Latency	122.7 ms

4. AWS Resource Analysis

Lambda: No Lambda throttling was observed during the test, and the function scaled successfully to handle the incoming request volume without any errors or delays caused by scaling limits.

DynamoDB: No read or write throttling occurred during the test. The table's capacity remained well within limits throughout the entire load test, indicating that the current provisioning is sufficient for this level of traffic.

Overall Service Performance: The application remained stable throughout the 60-second load test. All 3000 requests completed successfully with zero failures. Response times remained low and consistent, indicating good application performance under the tested load, with no noticeable service degradation.

5. Recommendations

To prepare the system for approximately 10× higher traffic, the following optimizations are recommended:

- Enable Provisioned Concurrency for Lambda to reduce cold starts.
- Enable API Gateway caching for frequently requested data.
- Configure DynamoDB Auto Scaling or On-Demand Capacity.
- Add CloudWatch alarms for latency, errors, and throttling.
- Optimize Lambda code to reduce execution time.
- Continue regular load testing with higher traffic levels.

6. Conclusion

The Employee Learning & Skill Certification Tracker successfully handled the simulated load, maintaining stable and consistent response times with zero request failures throughout the test. These results indicate that the system is capable of supporting moderate production workloads. Implementing the recommended optimizations will further improve the application's scalability and help it handle approximately 10 times higher traffic with continued reliability.